ECE
Probabilistic Systems Analysis

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Review

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}, \qquad \text{assuming } \mathbf{P}(B) > 0$$

- Multiplication rule:

$$\mathbf{P}(A \cap B) = \mathbf{P}(B) \cdot \mathbf{P}(A \mid B) = \mathbf{P}(A) \cdot \mathbf{P}(B \mid A)$$

- Total probability theorem:

$$\mathbf{P}(B) = \mathbf{P}(A)\mathbf{P}(B \mid A) + \mathbf{P}(A^c)\mathbf{P}(B \mid A^c)$$

- Bayes rule:

$$\mathbf{P}(A_i \mid B) = \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\mathbf{P}(B)}$$

Independence of Two Events

Definition of independence: $\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$

- **"Defn:"** $\mathbf{P}(B \mid A) = \mathbf{P}(B)$
  - "occurrence of A provides no information about B's occurrence"
- Recall that $\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B \mid A)$
- **Defn:** $\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$
  - Symmetric with respect to A and B
    - applies even if $\mathbf{P}(A) = 0$
    - implies $\mathbf{P}(A \mid B) = \mathbf{P}(A)$

A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. Let the events A, B, and C be defined as follows:

$$A = \{(1, b), (2, b)\}, \text{ "black ball selected";}$$

$$B = \{(2, b), (4, w)\}, \text{ "even-numbered ball selected"; and}$$

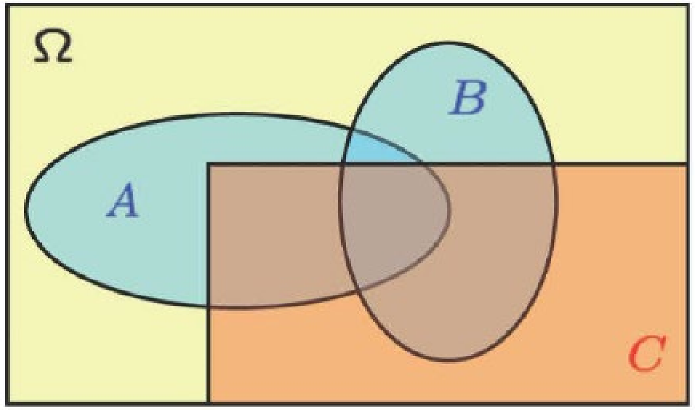
$$C = \{(3, w), (4, w)\}, \text{ "number of ball is greater than 2."}$$

Are events A and B independent? Are events A and C independent?

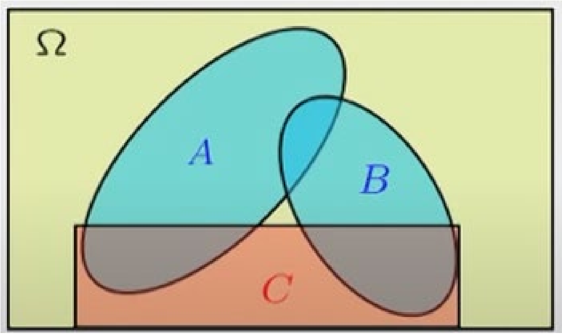
Concept	Meaning	Formulas
Disjoint	A and B cannot occur at the same time	$A \cap B = \emptyset$, $P(A \cup B) = P(A) + P(B)$
Independent	A does not give any information about B	$P(A\|B) = P(A), P(B\|A) = P(B)$ $P(A \cap B) = P(A)P(B)$

Conditional Independence

- Conditional independence, given C, is defined as independence under the probability law $\mathbf{P}(\cdot \mid C)$



Assume A and B are independent



If we are told that C occurred, are A and B independent?

A box contains two coins: a regular coin and one fake two-headed coin ($P(H) = 1$). I choose a coin at random and toss it twice. Define the following events.

- A= First coin toss results in an H.
- B= Second coin toss results in an H.
- C= Coin 1 (regular) has been selected.

Find $P(A|C), P(B|C), P(A \cap B|C), P(A), P(B)$, and $P(A \cap B)$. Note that A and B are NOT independent, but they are *conditionally* independent given C.

Principles of Counting

4 shirts
3 ties
2 jackets

- r stages
- n_i choices at stage i

Number of choices is:

Principles of Counting

- Number of license plates with 3 letters and 4 digits $=$
- ... if repetition is prohibited $=$

Permutation

- Order of selection matters
- Without replacement

Number of ways of ordering n elements is:

Number of subsets of $\{1, \ldots, n\}$ $=$

Example: Find the probability of a six roll die of a (six sided die) all give different number

- Number of outcomes that make the event happen:

- Number of elements in the sample space:

k-permutations

$$n(n-1)\cdots(n-k+1) = \frac{n(n-1)\cdots(n-k+1)(n-k)\cdots 2\cdot 1}{(n-k)\cdots 2\cdot 1}$$

Combination

- Order of selection is **NOT** matters
- Without replacement

$\binom{n}{k}$: number of k-element subsets of a given n-element set $= \boxed{\dfrac{n!}{k!(n-k)!}}$

constructing an **ordered** sequence of k **distinct** items:

Combination

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\binom{n}{n} =$$

$$\binom{n}{0} =$$

$$\sum_{k=0}^{n} \binom{n}{k} =$$

Binomial probabilities

Example:
You have n independent coin tosses. Suppose P(H) = P, find the probability that head shows K times with all passible sequences.

sequence of independent **Bernoulli trials**.

- $\mathbf{P}(HTTHHH) =$
- $\mathbf{P}(\text{particular sequence}) =$
- $\mathbf{P}(\text{particular } k\text{−head sequence})$

$\mathbf{P}(k \text{ heads}) =$

Coin tossing problem

- event B: 3 out of 10 tosses were "heads".
  - Given that B occurred, what is the (conditional) probability that the first 2 tosses were heads?

Partitions

Ten people have a potluck. Five people will be selected to bring a main dish, three people will bring drinks, and two people will bring dessert. How many ways they can be divided into these three groups?

$$\binom{10}{5}\binom{5}{3}\binom{2}{2} = \frac{10!}{5!5!} \cdot \frac{5!}{3!2!} \cdot \frac{2!}{2!0!} = \frac{10!}{5!3!2!}.$$

I roll a die 18 times. What is the probability that each number appears exactly 3 times.

$$\frac{18!}{3!3!3!3!3!3!}\left(\frac{1}{6}\right)^{18}$$

Example 1.30. A class consisting of 4 graduate and 12 undergraduate students is randomly divided into four groups of 4. What is the probability that each group includes a graduate student?

Summary of Counting Results

- Permutations of n objects: $n!$
- k-permutations of n objects: $n!/(n-k)!$
- Combinations of k out of n objects: $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$
- Partitions of n objects into r groups with the ith group having n_i objects:

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1!\, n_2! \cdots n_r!}.$$

(a) A coin is tossed twice and the sequence of heads and tails is noted. Let S be the sample space of this experiment. Find all subsets of S.

(b) A coin is tossed twice and the number of heads is noted. Let S? be the sample space of this experiment. Find all subsets of S'.

Q2/ A die is tossed twice and the number of dots facing up in each toss is counted and noted in the order of occurrence.

1- Find the sample space.

2- Find the set A corresponding to the event "number of dots in first toss is bigger than number of dots in second toss."

3- Find the set B corresponding to the event "number of dots in first toss is 6."

4- Does A subset of B or does B subset of A?

5- Find $A \cap B^C$.

A die is tossed and the number of dots facing up is noted.

(a) Find the probability of the elementary events under the assumption that all faces of the die are equally likely to be facing up after a toss.

(b) Find the probability of the events: $A = \{\text{more than 3 dots}\}$; $B = \{\text{odd number of dots}\}$.

(c) Find the probability of $A \cup B$, $A \cap B$, A^c.

2.77. A nonsymmetric binary communications channel is shown in Fig. P2.3. Assume the input is "0" with probability p and "1" with probability $1 - p$.

(a) Find the probability that the output is 0.

(b) Find the probability that the input was 0 given that the output is 1. Find the probability that the input is 1 given that the output is 1. Which input is more probable?

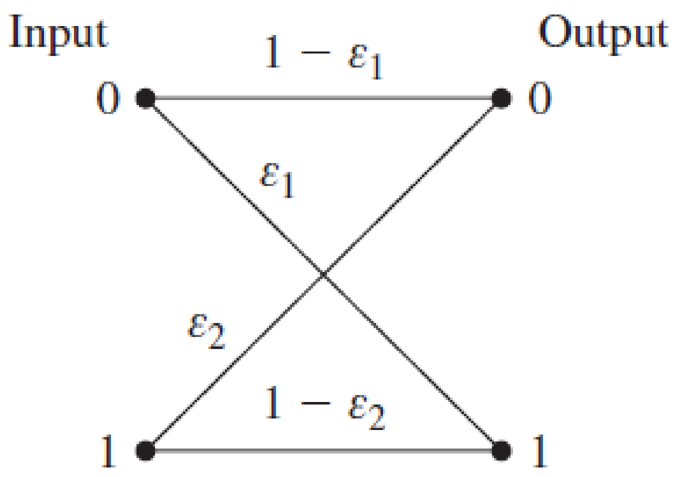


FIGURE P2.3